

Future Control Moves Memory: A Long-Horizon Moved-Bottleneck Diagnostic for Synthetic Agents

Jawaun Brown 2026-07-03

Abstract

This paper asks a narrow question about finite neural agents: when one early state variable becomes future-critical for a delayed decision, does the agent's memory geometry become specifically sensitive to that variable, and does that sensitivity move when the critical variable moves?

We introduce the **long-horizon moved-bottleneck diagnostic**. Four early clue slots are matched for salience and frequency. One slot is selected as the future-critical bottleneck, moved across registered positions, and queried only after a long delay. The primary metric is not just final accuracy; it is the relative displacement of the final hidden state under single-slot bit flips. The registered gate requires the final-state sensitivity peak to follow the moved critical slot while remaining absent in a visible-control condition.

The result is positive across a ladder of increasingly agent-like synthetic regimes. A transformer on Modal L4 solves the base delayed-memory task, survives longer horizons through sequence length 384, and preserves the moved bottleneck through closed-loop external-state handoff, repair after tool error, parsed structured calls, multi-field tool schemas, stochastic API failures, an 8-slot larger argument namespace, an alias-rich argument surface, parser-facing text arguments, fixed-length generated JSON strings, and autoregressively decoded JSON-like action strings. Across the confirmed Modal reports, the relevant bottleneck groups reach 1.000 closed-loop accuracy and positive moved-slot specificity, while visible controls keep specificity near zero.

The first prompt-level transfer produced a useful controlled strong negative at one hidden site: Qwen/Qwen2.5-0.5B-Instruct emits valid parser-scored JSON actions and passes the behavioral gates, but final-prompt hidden specificity is not confirmed. A follow-up localization sweep resolves that ambiguity. Across Qwen/Qwen2.5-0.5B-Instruct, Qwen/Qwen2.5-1.5B-Instruct, and HuggingFaceTB/SmolLM2-1.7B-Instruct, all behavior controls pass and 17 registered hidden sites pass. The strongest signal appears at final generated JSON tokens, with additional late/final prompt-token sites passing for Qwen 1.5B.

The allowed claim is deliberately modest: in this synthetic neural-agent setting, **future control relevance can move finite memory-state and tool-commitment sensitivity**. This is not a production-agent benchmark, production-agent benchmark or consciousness claim. Its value is as a cheap, reproducible diagnostic for whether an agent's internal state and generated action trajectory track the variables that will later control action.

1. Motivation

Many long-horizon agent evaluations ask whether an agent eventually gives the right answer. That is necessary but weak. A system can succeed by a shortcut, by storing everything uniformly, by using local query cues, or by relying on a privileged final token. The moved-bottleneck diagnostic tests a sharper property: whether the agent's internal memory metric follows the variable that will matter later.

This matters for three fields.

1. **Interpretability**: it gives a controlled way to ask where future-relevant information lives in a model's hidden state.
2. **Agent evaluation**: it tests delayed commitment and repair, not just answer selection at the final step.
3. **Safety and monitoring**: it suggests a cheap diagnostic for variables that silently become control bottlenecks in long-horizon systems.

The experiment is intentionally synthetic. That is a feature, not a defect, for this stage: the task isolates memory transport, critical-variable movement, and tool-interface hardening before introducing the confounds of natural language and real

APIs.

2. Diagnostic Design

Each episode contains matched early clues:

```
slot_0: bit b0
slot_1: bit b1
slot_2: bit b2
slot_3: bit b3
...
delayed query
```

In the `bottleneck` condition, the correct final action is the bit at one registered critical slot. The critical slot is moved across runs. In the `visible_control` condition, the final answer is visible at the query token and the early clues are matched distractors.

For each trained cell, the metric intervention flips one early clue bit at a time and measures the final hidden-state displacement. Slot densities are z-scored within model. The primary specificity statistic is:

```
z_density(critical_slot) - mean z_density(noncritical_slots)
```

The core gates are:

- behavior: bottleneck accuracy at least 0.90;
- metric transport: bottleneck memory-specificity CI lower bound above zero;
- rank: the critical slot ranks above chance among registered slots;
- visible-control null: visible-control specificity remains below 0.50.

The tool-interface regimes add gates for committed slot/value accuracy, schema validity, parsed action validity, repair behavior, stochastic failure handling, and no-op behavior after successful first calls.

3. Evidence Ladder

The experiment was advanced through a sequence of regimes. Each regime keeps the moved critical slot and visible-control null, then changes one surface of the agent's interaction with future-relevant information.

Regime	What changed	Cells	Key signal	Report key
Base moved bottleneck	Delayed query over four matched clue slots	64	Bottleneck final accuracy 1.000; memory specificity +2.309; visible null	base
Horizon stress	Delay lengths 128, 256, 384	96	All lengths preserve final accuracy 1.000 and moved-slot specificity	horizon
Closed-loop tool handoff	Model commits slot/value to recover external state	32	Tool bottleneck final accuracy 1.000; slot/value commitment 1.000	commit

Repair bottleneck	First tool attempt receives an error; model must repair	48	Repair condition final accuracy 1.000; repair slot/value 1.000	repair
Structured tool call	One parsed JSON-like action token replaces direct heads	48	Direct and repair structured calls pass schema and parsed-slot gates	structured
Multifield schema	Separate opcode, slot, and value fields	48	Multifield direct/repair groups pass field, schema, and parsed-value gates	multifield
Stochastic tool failure	First-call success/failure sampled per episode	32	Failed repair and success no-op both 1.000; failure rate 0.506	stochastic
8-slot stochastic	Larger argument namespace and longer sequence	64	8-slot stochastic gates pass; memory specificity +3.023	8slot
Alias argument surface	Three equivalent aliases per canonical slot	32	Alias parsed slot/value, failed repair, and success no-op all 1.000	alias
Text argument surface	Parser-facing phrases such as <code>second clue</code> replace alias IDs	32	Text parsed slot/value, failed repair, and success no-op all 1.000	text
Generated JSON surface	Model emits a fixed-length JSON-like token sequence	32	Generated sequence/schema, parsed slot/value, and repair gates all 1.000	generated_json
Autoregressive JSON surface	JSON action is decoded token-by-token from commit state	32	Greedy decoded sequence/schema, parsed slot/value, and repair gates all 1.000	autoregressive_json
Prompt JSON transfer	Pretrained open model emits parser-scored JSON text	64	Controls and behavior pass; hidden specificity CI crosses zero	prompt_json_transfer
Prompt JSON hidden localization	Multi-model, multi-layer, multi-token hidden-site grid	192 behavior cells + 576 hidden rows	Controls and behavior pass for three models; 17 hidden sites pass	prompt_json_hidden_localization
Prompt JSON fixed-action localization	Teacher-forced constant assistant actions under slot flips	192 behavior cells + 960 hidden rows	Controls and behavior pass; 24 hidden sites pass after generated action tokens are held fixed	prompt_json_fixed_action_localization

Prompt JSON causal patch	Patch base hidden state into critical-slot-flipped prompt before JSON value token	1,536 causal-patch rows	Value-prefix patch sites pass in all three model families; prompt-final sites do not pass	prompt_json_causal_patch
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All confirmed sweeps used Modal L4, not H100/H200. The timeout-based conservative spend guard for the listed confirmed reports is under USD 180 in aggregate; individual confirmed passes used guards of USD 1.08 to USD 17.26. Actual runtime was much lower than the timeout budget, with the latest autoregressive JSON pass averaging 15.08 seconds per remote cell. The prompt-level transfer used one L4 container for all 64 logical cells, with a conservative timeout guard of USD 1.08 and a 212.0 second remote runtime. The prompt-level hidden-localization sweep used three parallel L4 model jobs, 192 behavior cells, 576 hidden rows, and a conservative timeout guard of USD 6.47. The fixed-action localization counterfactual used the same three parallel L4 model jobs, 192 behavior cells, 960 hidden rows, and the same conservative timeout guard of USD 6.47. The causal-patch pass used three parallel L4 model jobs, 1,536 patch rows, and the same conservative timeout guard of USD 6.47.

Report keys map to committed summaries under `experiments/long_horizon_bottleneck/results/`: `base = modal_transformer_l4_8seed_2026_07_02.md`, `horizon = modal_transformer_l4_horizon_4seed_2026_07_02.md`, `commit = z_closed_loop_tool_commitment_l4_4seed_2026_07_02.md`, `repair = zz_tool_recovery_l4_4seed_2026_07_02.md`, `structured = zzz_structured_tool_call_l4_4seed_2026_07_03.md`, `multifield = zzzz_multifield_tool_schema_l4_4seed_2026_07_03.md`, `stochastic = zzzzz_stochastic_tool_failure_l4_4seed_2026_07_03.md`, `8slot = zzzzzz_stochastic_tool_failure_8slot_l4_4seed_2026_07_03.md`, `and alias = zzzzzzz_alias_argument_surface_l4_4seed_2026_07_03.md`, `text = zzzzzzzz_text_argument_surface_l4_4seed_2026_07_03.md`, `generated_json = zzzzzzzzz_generated_json_surface_l4_4seed_2026_07_03.md`, `and autoregressive_json = zzzzzzzzzz_autoregressive_json_surface_l4_4seed_2026_07_03.md`, `and prompt_json_transfer = zzzzzzzzzzz_prompt_json_transfer_l4_4seed_2026_07_03.md`, `and prompt_json_hidden_localization = zzzzzzzzzzzz_prompt_json_hidden_localization_l4_4seed_2026_07_03.md`, `and prompt_json_fixed_action_localization = zzzzzzzzzzzzz_prompt_json_fixed_action_localization_l4_4seed_2026_07_03.md`, `and prompt_json_causal_patch = zzzzzzzzzzzzzz_prompt_json_causal_patch_l4_4seed_2026_07_03.md`.

4. Prompt-Level JSON Transfer and Hidden Localization

The autoregressive JSON pass completed the synthetic ladder: the trained model could decode parser-scored JSON-like action strings token by token under stochastic first-call failure and conditional repair. The next question was whether the same diagnostic transfers to a pretrained prompt-level model.

The prompt-level transfer used `Qwen/Qwen2.5-0.5B-Instruct` on one Modal L4 container. The model saw natural-language memory records and had to emit one JSON object:

```
{"tool": "read_slot", "slot": "second clue", "value": 0}
```

or:

```
{"tool": "noop"}
```

The confirmatory run used 64 logical cells: 4 conditions, 4 moved critical slots, and 4 seeds, with 8 episodes per cell. Format, visible-control, and short-horizon controls passed. The behavioral bottleneck also passed: closed-loop final accuracy was 0.977, first parsed slot/value were 0.984/0.977, failed-repair slot/value were 1.000/1.000, and success repair no-op was 1.000.

The full prompt-level gate nevertheless failed by the preregistered hidden metric. Critical-slot memory specificity was

+0.695, but its 95% bootstrap interval was [-0.376, 2.080], so the lower bound was not above zero. The rank metric was above chance by mean rank percentile (0.594), but the primary specificity gate did not pass.

This is a controlled strong negative, not an interface failure. The model can solve the parser-scored prompt-level moved-bottleneck behavior, including stochastic repair, but this run does not show reliable hidden-state transport of the future-critical slot.

The hidden-localization follow-up asked whether that negative was specific to the measured site. It ran three cheap open models in parallel on Modal L4: Qwen/Qwen2.5-0.5B-Instruct, Qwen/Qwen2.5-1.5B-Instruct, and HuggingFaceTB/SmolLM2-1.7B-Instruct. The sweep preserved the same behavior conditions, then measured hidden specificity at `prompt_final`, `generated_first`, and `generated_final` token positions across early, mid, late, and final layer aliases.

This run was positive. All behavior controls passed for all three models. Behavioral bottleneck accuracy was 1.000 for SmolLM2 1.7B, 0.961 for Qwen 0.5B, and 0.938 for Qwen 1.5B. Seventeen registered hidden sites passed both the specificity and rank gates. The strongest sites were generated-final states: SmolLM2 1.7B generated-final/mid had specificity +228.301 with CI [129.438, 386.540]; Qwen 1.5B generated-final/early had +216.902 with CI [119.175, 332.137]; and Qwen 0.5B generated-final/early had +183.444 with CI [99.032, 279.601]. Qwen 1.5B also passed late/final prompt-final sites.

The interpretation is therefore narrower and stronger than the first prompt run: final behavior transfers, and hidden localization is present in the generated action trajectory. The original negative was a measurement-site failure, not evidence that prompt-level hidden geometry is absent.

The fixed-action counterfactual then tested the main confound in that positive result. For each base prompt and every slot-flipped prompt, the assistant action tokens were teacher-forced to remain fixed: either `{"tool": "noop"}` or `{"tool": "read_slot", "slot": "<critical slot phrase>", "value": 0}`. The model therefore could not reveal the moved slot merely by generating a different answer string under the counterfactual prompt.

This stricter run was also positive. It used the same three-model L4 grid, 192 behavior rows, and 960 hidden rows across `prompt_final`, `fixed_noop_first/final`, and `fixed_read_first/final` positions. Behavior again passed for all three models. Twenty-four registered hidden sites passed both hidden gates. Fixed noop final-layer sites passed in all three model families, with the strongest at Qwen 1.5B `fixed_noop_final/final` (specificity +12.514, CI [8.035, 18.310], rank 0.984). This removes the simple generated-answer-token explanation for the localization effect.

Finally, the causal-patch pass asked whether the localized state can shift a behavior-adjacent readout. The runner appended a partial assistant action prefix ending immediately before the JSON value token, patched one donor hidden state from the base prompt/prefix into the critical-slot-flipped prompt/prefix, and measured the donor-value minus corrupted-value next-token logit margin.

This run was positive across all three model families. Clean prompts preferred the donor value, corrupted prompts preferred the flipped value, and `value_prefix_final` late/final patch sites passed in every model. The strongest site was SmolLM2 `value_prefix_final/final`, with patch effect +7.053 and CI [6.738, 7.344]; Qwen 0.5B final had +5.503 [5.299, 5.697], and Qwen 1.5B final had +6.916 [6.375, 7.453]. Prompt-final patch sites did not pass. That negative control localizes the causal leverage to the action/value prefix state rather than the raw final prompt token.

5. Interpretation

The core result is a **metric transport** result. The future-critical variable does not merely determine the final label; it becomes the slot to which the agent's final hidden state and commitment heads are specifically sensitive. When the critical slot moves, the sensitivity peak moves with it. When the final answer is visible at query time, the early-slot peak disappears.

The tool regimes add a second claim: the moved bottleneck survives when the agent has to externalize the critical variable through a tool interface. It must commit an argument, receive or fail to receive an external return, repair when needed, no-op when not needed, and still preserve the correct delayed decision.

This makes the diagnostic more relevant to long-horizon agents than to ordinary sequence classifiers. The bottleneck is not just "remember bit 2." It becomes "know which early variable must later be committed through an interface, recover it under feedback, and ignore matched distractors."

The prompt-level results sharpen the interpretation. Final behavior alone is not enough: a small open model can pass the prompt-level JSON behavior while one registered hidden site fails. But localization across token positions and layers recovers a positive signal, and the fixed-action counterfactual shows that the signal can survive even when generated answer tokens are held constant. The causal-patch pass then shows that the value-prefix state can shift the behavior-adjacent value-token readout. That split is the point of the diagnostic: it can distinguish interface behavior, prompt-state memory, action-surface commitment, and causal leverage over the next action token.

6. Boundaries

The strongest honest statement is:

Future control relevance can move finite memory and commitment sensitivity in a synthetic long-horizon neural agent, and that moved bottleneck survives progressively harder parsed tool-interface regimes.

The result does not establish:

- production API reliability;
- autonomous language-agent tool use;
- robustness across natural-language prompt families;
- multi-step planning in an open environment;
- human cognition or consciousness.

The latest prompt-level result uses a real pretrained tokenizer and teacher-forced/generated JSON text, but it is still a compact harness, not an autonomous agent or real API environment.

7. Why This Is Valuable

The experiment's field value is not that it solves a benchmark. It gives a small, inspectable test for a pattern that current agent evaluations often miss: future relevance can reorganize internal memory geometry and action commitment.

A useful follow-on benchmark could ask, for a real agent or language model:

1. Which early facts become future-critical under a delayed objective?
2. Do internal representations or tool-call logits become specifically sensitive to those facts?
3. Does that specificity move when the objective moves?
4. Does it remain absent when the answer is visible later?
5. Does it survive tool failure, repair, and aliasing of argument names?

That is a sharper evaluation target than final-task success alone.

8. Remaining Work

The synthetic mechanism ladder is complete enough to write up and share. The prompt-level hidden-localization replication is positive, and the fixed-action counterfactual removes the simplest generated-token identity confound. The fixed-prefix causal patch also shows behavior-adjacent causal leverage at the JSON value-token readout. The next regimes answer different questions:

- **Prompt-level robustness:** rerun the frozen gate across prompt families, longer contexts, and API models to test whether fixed-action localization is robust rather than prompt-specific.
- **Stronger causal mediation:** patch multi-token generation or run path patching to test whether the measured state mediates the full tool action, not only the next value token.
- **LLM-agent transfer:** add longer tool contexts, natural-language argument aliases, and API-style parser recovery

around the prompt-level version.

- **Multi-step planning:** require two or more future commitments where the critical bottleneck changes after intermediate feedback.
- **Interpretability probes:** compare the hidden-state metric to attention, activation patching, or linear probes over the critical slot.

The most valuable immediate next step is to publish the diagnostic as a compact benchmark card: synthetic ladder positive, prompt-level behavior positive, prompt-level hidden localization positive, and fixed-action localization positive after generated action tokens are held constant, with a first causal patch result showing value-token logit control.

9. Reproducibility

Experiment code lives in `experiments/long_horizon_bottleneck/`.

Synthetic ladder runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/long_horizon_bottleneck/modal_stochastic_tool_failure_sweep.py \
  --seeds 4 --train-steps 900 --architectures transformer \
  --conditions autoregressive_json_bottleneck,autoregressive_json_visible_control \
  --critical-slots 0,1,2,3 \
  --aliases-per-slot 3 \
  --failure-probability 0.5 \
  --budget-usd 25 \
  --out artifacts/long_horizon_bottleneck/autoregressive_json_surface_l4.json
```

Prompt-level confirmatory runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/long_horizon_bottleneck/modal_prompt_json_transfer_sweep.py \
  --model-id Qwen/Qwen2.5-0.5B-Instruct \
  --seeds 4 \
  --episodes-per-cell 8 \
  --hidden-metric-episodes 2 \
  --critical-slots 0,1,2,3 \
  --budget-usd 25 \
  --base-seed 20260800 \
  --out artifacts/long_horizon_bottleneck/prompt_json_transfer_l4.json
```

Prompt-level hidden-localization runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/long_horizon_bottleneck/modal_prompt_json_hidden_localization_sweep.py \
  --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-Instruct \
  --seeds 4 \
  --episodes-per-cell 8 \
  --hidden-metric-episodes 2 \
  --critical-slots 0,1,2,3 \
  --hidden-positions prompt_final,generated_first,generated_final \
  --hidden-layers early,mid,late,final \
  --budget-usd 25 \
  --base-seed 20260850 \
  --out artifacts/long_horizon_bottleneck/prompt_json_hidden_localization_l4.json
```

Prompt-level fixed-action localization runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/long_horizon_bottleneck/modal_prompt_json_hidden_localization_sweep.py \
  --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-
Instruct \
  --seeds 4 \
  --episodes-per-cell 8 \
  --hidden-metric-episodes 2 \
  --critical-slots 0,1,2,3 \
  --hidden-positions
prompt_final,fixed_noop_first,fixed_noop_final,fixed_read_first,fixed_read_final \
  --hidden-layers early,mid,late,final \
  --budget-usd 25 \
  --base-seed 20260900 \
  --out artifacts/long_horizon_bottleneck/prompt_json_fixed_action_localization_l4.json
```

Prompt-level fixed-prefix causal patch runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/long_horizon_bottleneck/modal_prompt_json_causal_patch_sweep.py \
  --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-
Instruct \
  --seeds 4 \
  --episodes-per-cell 8 \
  --critical-slots 0,1,2,3 \
  --patch-positions prompt_final,value_prefix_final \
  --patch-layers late,final \
  --budget-usd 25 \
  --base-seed 20260950 \
  --out artifacts/long_horizon_bottleneck/prompt_json_causal_patch_l4.json
```

Local verification for the code paths:

```
uvx ruff check experiments/long_horizon_bottleneck tests/test_long_horizon_bottleneck.py
uvx ty check experiments/long_horizon_bottleneck tests/test_long_horizon_bottleneck.py
python -m compileall -q experiments/long_horizon_bottleneck tests/test_long_horizon_bottleneck.py
python -m pytest tests/test_long_horizon_bottleneck.py -q
```

The raw Modal artifacts are kept under gitignored `artifacts/`; committed result reports in `experiments/long_horizon_bottleneck/results/` summarize every run.